

Proof. If $\bar{\lambda} \notin \sigma(D|_{\mathcal{J}})$ the resolvent $R = (D - \bar{\lambda}I|_{\mathcal{J}})^{-1}$ has the form

$$Rx = \ell(x)\phi_{\bar{\lambda}} + V(I - \bar{\lambda}V)^{-1}x,$$

with ℓ a continuous linear functional on $C^\infty(D)$. Since $\mathcal{W}\varphi(\lambda) = 0$ it follows that

$$\varphi_\lambda(x) = \varphi(V(I - \bar{\lambda}V)^{-1}x) = \varphi(Rx) = 0,$$

for all $x \in \mathcal{J}$. By the form of φ it follows easily that φ_λ has order $N - 1$ if $N > 0$. Moreover,

$$\mathcal{W}\varphi_\lambda(z) = \varphi(V(I - \bar{\lambda}V)^{-1}\phi_z)^\# = \varphi((z - \bar{\lambda})^{-1}(\phi_z - \phi_{\bar{\lambda}}))^\# = \frac{\mathcal{W}\varphi(z)}{z - \bar{\lambda}}.$$

(ii) follows by a repeated application of (i). $\square$

6.3 Residual subspaces and their annihilators

As pointed out before, we are especially interested in D -invariant subspaces $\mathcal{J}$ with $\sigma(D|_{\mathcal{J}}) = \emptyset$. Such subspaces will be called *residual*. They possess an alternative characterization based on the following simple observation.

Lemma 3. *If $\mathcal{J} \in \mathcal{J}$ and p is a polynomial then $p(D)\mathcal{J}$ is closed.*

Proof. It suffices to show that $(D - \lambda I)\mathcal{J}$, $\lambda \in \mathbb{C}$ is closed. Let (x_k) be a sequence in $\mathcal{J}$ such that $(D - \lambda I)x_k \rightarrow y \in H$. Then there is a sequence (c_k) in $\mathbb{C}$ such that

$$V(I - \lambda V)^{-1}(D - \lambda I)x_k = x_k - c_k\phi_\lambda \rightarrow V(I - \lambda V)^{-1}y, \quad (30)$$

when $k \rightarrow \infty$. If $\phi_\lambda \in \mathcal{J}$, it follows that (x_k) converges to $x \in \mathcal{J}$ and $y = (D - \lambda I)x \in (D - \lambda I)\mathcal{J}$. If $\phi_\lambda \notin \mathcal{J}$, let φ be a continuous linear functional in $\mathcal{J}^\perp$ with $\varphi(\phi_\lambda) = 1$. Then applying φ to both sides of (30) we obtain that (c_k) converges and as above, we obtain that (x_k) converges to $x \in \mathcal{J}$ and $y = (D - \lambda I)x \in (D - \lambda I)\mathcal{J}$. $\square$

With the lemma in hand, the characterization of residual subspaces is as follows.

Proposition 12. *A subspace $\mathcal{J} \in \mathcal{J}$ is residual if and only if*

$$\mathcal{J} = \bigcap \{p(D)\mathcal{J} : p \text{ polynomial}\}.$$

Proof. If $\sigma(D|_{\mathcal{J}}) = \emptyset$ we have $(D - \lambda)\mathcal{J} = \mathcal{J}$ for all λ and the equality in the statement holds. Conversely, if the equality holds, then $(D - \lambda I)\mathcal{J} = \mathcal{J}$ for each λ . Suppose for a contradiction that $D - \lambda_0 I$ is not injective for some λ_0 . Then for each $j \geq 0$ there exists a nonzero $x_j \in \mathcal{J}$, such that $(D - \lambda_0)^j x_j = \phi_{\lambda_0}$. From this it follows that $V_{\lambda_0}^j \phi_{\lambda_0} \in \mathcal{J}$ for all $j \geq 0$, and $\mathcal{J} = C^\infty(D)$ by Proposition 8 (ii). $\square$

Note that for every subspace $\mathcal{J} \in \mathcal{J}$ we can consider its *residual part*

$$\mathcal{J}_{res} = \bigcap \{p(D)\mathcal{J} : p \text{ polynomial}\}.$$

By the above proposition this is a closed residual subspace of $\mathcal{J}$

Let us now turn to annihilators of residual subspaces. It turns out that these spaces can be described using the generalized Fourier transform $\mathcal{W}$.

To state our result we use the same notation as in Theorem 8, that is $\mathcal{W}H = e^{-i\alpha z}\mathcal{H}(E)$, with $\alpha \in \mathbb{R}$.